

GIRINATH M

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student passionate about Embedded Systems, PCB Design, IoT and Electronics Hardware. Hands-on experience with ESP32, EasyEDA, Arduino and sensor-based projects through internships and academic work.

EDUCATION

Bachelor of Degree - Electronics and Communication Engineering. CGPA: 8.5% **2024-2028**
NPR College of Engineering and Technology, Natham, Dindigul.

HSC-M.S.P. Solai Nadar Memorial Higher Secondary School, Dindigul. Percentage: 68.83% **2023-2024**

SSLC- Kalaimagal Higher Secondary School,Vadamadurai, Dindigul. Percentage: 64.4% **2021-2022**

EXPERIENCE

EMBEDDED & IoT Intern | Z00L00P

20 Apr 2026 - Present

- Designed and validated PCB schematics and layouts using EasyEDA in accordance with industry design standards.
- Evaluated electronic component datasheets to support accurate component selection and hardware design decisions. Conducted product research and contributed to Research & Development (R&D) activities for embedded and IoT-based solutions.

IOT INTERN | TARCIN ROBOTIC LLP

15 Dec 2025 - 07 Jan 2026

- Developed an IoT-based radar detection system for real-time object detection and monitoring.
- Integrated sensors, microcontroller and wireless communication for accurate data transmission.
- Performed hardware integration, testing and system optimization to improve reliability and performance.

TECHNICAL SKILLS

Hardware: Arduino , Esp32 | **PCB Design :** EasyEDA , KiCad , Proteus | **Tools :** Thingspeak , Github , VS Code | **Programming Language :** C , C++ , Python (Basics) | **Certification:** NIELIT - VLSI for Beginners , Introduction for MATLAB & Simulink , Embedded for Beginners | **Language:** Tamil , English

PROJECTS

AIR QUALITY MONITORING SYSTEM

FEB'25

Developed an Arduino Uno-based air quality monitoring system using MQ-2, MQ-9, MQ-135 gas sensors and DHT11 sensor. Integrated an I2C LCD display for real-time monitoring of environmental parameters. Performed sensor interfacing, hardware integration and system testing.

Radar Detection System

DEC'25

Developed an Arduino Uno-based radar detection system using an ultrasonic sensor and servo motor for real-time object detection. Designed a web interface to visualize detection data and improve user interaction. Integrated hardware components, sensor interfacing and embedded programming for accurate operation.

IoT-Based Crop Root Health Monitoring System

APR'26

Developed an ESP32-based IoT system to monitor crop root health using DHT11, capacitive Soil moisture, DS18B20 and TDS sensors. Collected and processed real-time soil and environmental data for continuous monitoring. Integrated ThingSpeak for cloud-based data storage and visualization. Developed a web application for monitoring of live sensor data